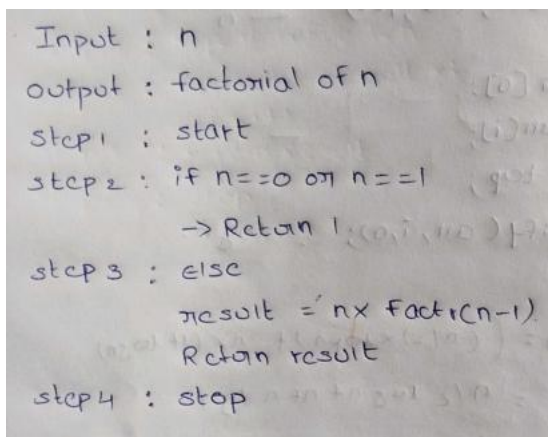


Experiment - 4

Aim: Implementation and Time analysis of factorial program using iterative and recursive method.

1)Recursive method

Algorithm:



```
Input : n
output : factorial of n
step 1 : start
step 2 : if n==0 or n==1
    -> Return 1
step 3 : else
    result = n x Fact(n-1)
    Return result
step 4 : stop
```

Program:

```
#include <iostream>

using namespace std;

class Factorial

{

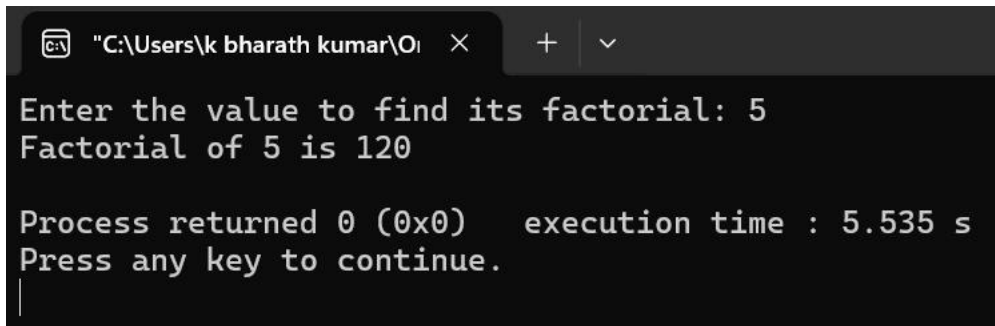
public:

    int factr(int n)
```

```
{  
  
    if (n == 0 || n == 1)  
  
    {  
  
        return 1;  
  
    }  
  
    else  
  
    {  
  
        int result = n * factr(n - 1);  
  
        return result;  
  
    }  
  
}  
  
};  
  
int main()  
  
{  
  
    int n, result;  
  
    Factorial obj;  
  
    cout << "Enter the value to find its factorial: ";  
  
    cin >> n;
```

```
result = obj.factr(n);  
  
cout << "Factorial of " << n << " is " << result << endl;  
  
return 0;  
  
}
```

Output:



```
"C:\Users\k bharath kumar\O:  X  +  v  
Enter the value to find its factorial: 5  
Factorial of 5 is 120  
  
Process returned 0 (0x0)   execution time : 5.535 s  
Press any key to continue.  
|
```

Time complexity:

```

int fact(int n)
{
    if (n == 0 || n == 1) — n
    {
        return 1; — 1
    }
    else
    {
        int result = n * fact(n-1); — n-1
        return result; — n-1
    }
}

```

$\Rightarrow T(n) = n + 1 + (n-1) + (n-1) + \dots + 1$
 $= 3n - 1$
 $= O(n)$

Best case $\rightarrow$ Ex: $n=1$

```

int fact(1)
{
    if (1 == 0 || 1 == 1) — 1
    {
        return 1; — 1
    }
}

```

$\Rightarrow T(1) = 1 + 1 = 2 = O(1)$
Best case $T(n) = O(1)$

Average case $\rightarrow$ Ex: $n=3$

calculate: $\text{fact}(3) \rightarrow \text{fact}(2) \rightarrow \text{fact}(1)$

```

if (n == 0 || n == 1) — n
result = n * fact(n-1) — n-1
return result — n-1
return 1 — 1

```

$\Rightarrow n + n-1 + n-1 + 1 = 3n - 1 = O(n)$
Average case $T(n) = O(n)$

Worst case Ex: $n=7$

calculate: $f(7) \rightarrow f(6) \rightarrow f(5) \rightarrow f(4) \rightarrow f(3) \rightarrow f(2) \rightarrow f(1)$

```

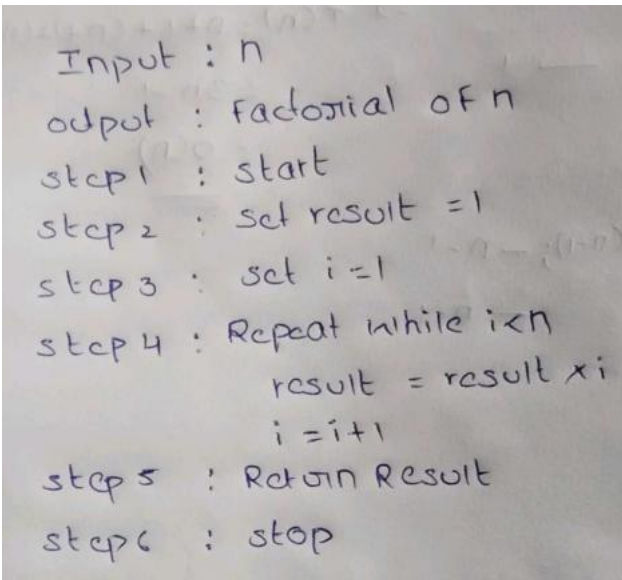
if (n == 0 || n == 1) — n
result = n * fact(n-1) — n-1
return result — n-1
return 1 — 1

```

$\Rightarrow n + n-1 + n-1 + \dots + 1 = 3n - 1 = O(n)$
Worst case $T(n) = O(n)$

2) Iterative Method

Algorithm:



```
Input : n
Output : Factorial of n
step 1 : start
step 2 : set result = 1
step 3 : set i = 1
step 4 : Repeat while i <= n
         result = result * i
         i = i + 1
step 5 : Return Result
step 6 : stop
```

Program:

```
#include <iostream>

using namespace std;

class Factorial

{

public:

    int fact(int n)

    {

        int result = 1;
```

```
for (int i = 1; i <= n; i++)  
  
    {  
  
        result = result * i;  
  
    }  
  
    return result;  
  
}  
  
};  
  
int main()  
  
{  
  
    int n,result;  
  
    Factorial obj;  
  
    cout << "Enter the value to find its factorial: ";  
  
    cin >> n;  
  
    result = obj.fact(n);  
  
    cout << "Factorial of " << n << " is " << result << endl;  
  
    return 0;  
  
}
```

Output:

```
"C:\Users\k bharath kumar\O... x + v
Enter the value to find its factorial: 7
Factorial of 7 is 5040

Process returned 0 (0x0)    execution time : 3.285 s
Press any key to continue.
```

Time complexity:

```
int fact (int n)
{
    int result = 1;
    for (int i=1; i<n; i++) — n+1
    {
        result = result * i; — n
    }
    return result; — 1
}

Best case → ex: if n=0
int fact (0)
{
    int result = 1;
    for (int i=1; i<0; i++) — 1
    {
        result = result * i; — 0
    }
    return result; — 1
}
Best case T(n) = O(1)

Average case → ex if n=3
⇒ fact (3)
it will run n+1 times
for (int i=1; i<3; i++) — n+1
{
    result = result * i; — n
}
⇒ T(n) = 2n + 3
⇒ O(n) — Average case
T(n) = O(n)

Worst case → ex if n=5
⇒ fact (5)
int result = 1;
for (int i=1; i<5; i++) — n+1
{
    result = result * i; — n
}
return result; — 1
Worst case T(n) = O(n)
```

Conclusion:

As per the experiment to Implementation and Time analysis of factorial program using iterative and recursive method., Output was verified successfully according to the Experiment.